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RN-SMOTE: Reduced Noise SMOTE based on DBSCAN for enhancing imbalanced data classification

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ABSTRACT

Machine learning classifiers perform well on balanced datasets. Unfortunately, a lot of the real-world data sets are naturally imbalanced. So, imbalanced classification is a serious problem in machine learning. The imbalanced class distribution misleads classifiers from correctly classifying the minor class. This paper introduces Reduced Noise-SMOTE (**RN-SMOTE**) for pre-processing imbalanced data. **RN-SMOTE** firstly, oversamples the training data using SMOTE which introduces noisy oversampled synthetic instances in the minority class. Then, applying DBSCAN to detect and remove noise. Next, the clean artificial instances are combined with the original data. Finally, **RN-SMOTE** applies SMOTE again to rebalance the dataset before introducing it to the underlying classifier. **RN-SMOTE** is evaluated using 9 different classifiers and 9 different imbalanced datasets with different imbalance ratios and five of them are used for outlier detection. The results proved that the performance of the classifiers has been improved with **RN-SMOTE** and outperformed the performance with original data and SMOTE with percentage based on the classifier, dataset and evaluation metric. Also, performance of **RN-SMOTE** has been compared to the performance of the current state of art and resulted in an increase up to 37.41%, 23.28%, 13.95% and 9.07% in terms of Recall, F1, Precision and Accuracy for **RN-SMOTE**.

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1. Introduction

Imbalanced datasets are those in which the number of samples in the different classes are greatly different (López et al., 2013). This imbalance in the class distribution is a challenging problem that has a bad effect on the performance of machine learning classifiers. So, the sensitivity of many machine learning classifiers to the class imbalance has been handled in many works of literature to improve the performance of these classifiers with imbalanced datasets (Batuwita and Palade, 2013; Boonchuay et al., 2017; Cheng et al., 2021; García et al., 2008). Really, class imbalance is not the only reason for the negative effect on classifiers when classifying imbalanced data, but some other reasons such as class

overlapping and the sample size should be considered (Sun et al., 2009).

There are two types of imbalances in real-world datasets, intrinsic and extrinsic. Intrinsic imbalance is an inherent property of the given data space (Haibo He and Garcia, 2009). On the contrary, the extrinsic imbalance is not a nature of the data space but results from economic, time, or storage reasons (Stefanowski, 2016). So, collecting new examples is not always the proper solution to the imbalance problem especially if the imbalance is the nature of the data. For example, the number of healthy people always exceeds greatly the number of patients in medical data which means an intrinsic imbalance in the data. In addition, most of the real-world datasets suffer from the presence of noise that affects the performance of classifiers on the given dataset (García et al., 2013; Hasan and Chu, 2022). This noise has a greater effect on minor samples classification compared to major samples (Weiss, 2004). So, noise reduction has a notable impact on the performance of classifiers when classifying imbalanced datasets. This impact will increase especially when balancing the data set using SMOTE which adds a random component and generates noise in the resulted balanced data.

All proposed approaches to handle the class imbalance problem are categorized as data-level, algorithm-level and hybrid

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approaches (García and Herrera, 2009). Also, these approaches are categorized as internal approaches referring to the algorithms and external approaches referring to data pre-processing (Cateni et al., 2014).

In algorithm-level approaches, the used classifier is modified to correctly classify the minority class samples and attenuate the sensitivity to the class imbalance (Liu, 2021). This includes the one class classifiers where the classifier is learned only from a single class and improves the performance for the minor class (Bellinger et al., 2012). For example, One Class Support Vector Machine (OCSVM) modifies the classifier bias towards the minor samples by adding a penalty. This method is referred to cost-sensitive approach (Krawczyk, 2016). Other classifiers use a threshold to determine the degree to which each sample is a member of a class (Kotsiantis et al., 2006).

Data-level approaches are independent of the used classifier where only the imbalanced dataset is modified or resampled producing a new balanced dataset that allows the used classifier to classify it correctly (López et al., 2012). Data-level approaches include resampling the data by adding more samples to the minor class (oversampling) or by removing samples from the major class (under sampling). Many well-known approaches are used for oversampling imbalanced datasets including Random Oversampling (ROS), Synthetic Minority Over-sampling Technique (SMOTE) with its different variations that overcome the limitations of ROS. Also, imbalanced datasets can be under sampled randomly through the use of well-known Random Under Sampling (RUS), or based on one of the popular informative methods, for example, but not limited to removing Tomek's Links or by using Neighbourhood Cleaning Rule. Also, the imbalanced dataset can be resampled by approaches that combine both of oversampling and under sampling such as Smote Tomek that combine both SMOTE as an oversampling technique and Tomek's Links as an under-sampling technique. Data-level approaches also includes the feature selection (Ali et al., 2013) where only a subset of all the features is used instead of using all given features. This is required with datasets with curse dimensionality where other techniques such as resampling and algorithm-level techniques are not capable of overcoming the imbalance problem (Wasikowski and Chen, 2010).

Hybrid approaches, combine both the data-level and algorithm-level approaches to gain advantages of both approaches. It includes the ensemble methods such as bagging and boosting ensemble classifiers (Tanha et al., 2020). Also, this category includes strategies to handle the imbalanced dataset from different perspectives such as noise reduction (Johnson and Khoshgoftaar, 2019).

This paper introduces a new model for handling the class imbalance problem. The proposed Reduced Noise-SMOTE (RN-SMOTE) applies the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) for detection of SMOTE noise then removing it from the training dataset. After that, the dataset is resampled using SMOTE for balancing it.

The main contributions of this paper are the following:

- Introduce a new pre-processing technique for imbalanced data which is RN-SMOTE by utilizing DBSCAN for noise removal after oversampling with SMOTE.
- Evaluate the proposed RN-SMOTE using different classifiers and different noisy datasets with different imbalance ratios and dimensionalities under different metrics.
- Compare the performance of the classifiers combined with **RN-SMOTE** with their performance over both SMOTE and original data.
- Compare the performance of **RN-SMOTE** with the performance of the current state of art.

The rest of this paper is organized as follows: Section 2 reviews the related work. Section 3 explains the proposed model by firstly introducing both the SMOTE and DBSCAN then describing **RN-SMOTE** in details. Section 4 shows the experiments and discusses the obtained results. And lastly, Section 5 is the conclusion and future work.

2. Related work

One of the effective solutions for imbalanced datasets classification is to change the problem scope and analyze the data from anomaly or noise detection point of view. This is because the original data may contain anomalies or noise. In addition, some resampling techniques such as SMOTE may generate additional noise which makes it worse. High noise in training datasets negatively affects the performance of the underlying classifier. This point of view for imbalanced data has been handled in many studies.

Recently, Asniar et al. (2021) introduced SMOTE-LOF for this purpose. In their proposal, SMOTE is used first for oversampling the imbalanced data then Local Outlier Factor (LOF) is used to detect the noise in the minor class in the resulted oversampled data which has been balanced. SMOTE-LOF had been evaluated with PIMA ("<https://www.kaggle.com/kumargh/pimaindiansdiabetescsv>," n.d.), Haberman ("<https://www.kaggle.com/saguneshgrover/haberman>," n.d.) and Glass ("<https://sci2s.ugr.es/keel/imbalanced.php#sub2A>," n.d.) datasets using Naïve Bayes (NB), Support vector machine (SVM) and C4.5 decision tree classifiers using WEKA tool. The performance of the used classifiers is compared with SMOTE-LOF, SMOTE and Original data using metrics including Accuracy, Precision, Recall, and F1 scores. SMOTE-LOF is evaluated only using three real-world datasets with a maximum number of attributes equaling 9 and a minimum imbalance ratio equal (81:225). Also, only three classifiers only are used for its evaluation. So, its performance is unknown on other types of classifiers and other real-world datasets with lower imbalance ratios and higher dimensionality. SMOTE-LOF utilizes LOF for noise detection. Unfortunately, LOF has a higher time complexity, estimated by $O(n^2)$ compared to other techniques such as DBSCAN which has a time complexity estimated by $O(n \log n)$ (Behera and Rani, 2016).

Nnamoko and Korkontzelos (2020) introduced a two-step pre-processing model for handling the class imbalance in diabetes prediction. In their proposal, the outliers are first searched in the imbalanced dataset using the interquartile range (IQR) algorithm. Then, these outliers are randomly oversampled with replacement and added to the original data. Finally, they applied SMOTE to obtain balanced data. Also, based on this sequence of steps, A. Banerjee et al. in Banerjee et al. (2021) introduced a Fused Oversampling Framework by addressing Outliers (FOFO) model in which outliers are also identified using IQR then oversampled using SMOTE and added to the original data. At last, the data is balanced using SMOTE. These two proposals are based on IQR for outlier detection which does not consider the number of the samples in the underlying dataset (Barbato et al., 2011). Also, their performance on other datasets with lower imbalance ratios is unexplained.

In (Revathi and Ramyachitra, 2021), Revathi et al. introduced a new combination between oversampling and noise reduction techniques which is Modified Borderline SMOTE with Noise Reduction (NRBSID). Their proposal first determines the propensity scores for all samples. Then, based on a threshold, noise samples are removed from the minor class. Finally, the cleaned data is oversampled using Borderline SMOTE. This proposal is based on

Borderline-SMOTE which improves the classifier performance for the minor class only while the major class is ignored (Bunkhumpornpat et al., 2011). Also, the maximum number of features for the used datasets is 13 attributes only. So, its performance for higher dimensionalities is unexamined.

In (Liang et al., 2020), Liang et al. introduced LR-SMOTE in which SVM and k-means are used to identify and eliminate the noise in the training data. Then, based on SMOTE, the data is oversampled by generating new samples which filtered based on the nearest neighbours. This proposal was evaluated only using two classifiers which are SVM and RF. So, its performance on other classifiers with different operation theories is unrecognized. Also, the minimum imbalance ratio for the datasets used in the evaluation is (42: 689) and the maximum number of attributes is 14 only. So, its performance on other datasets with lower imbalance ratios or with higher dimensionalities is unillustrated.

Outlier Detection-Based Oversampling Technique (ODBOT) (Ibrahim, 2021) is another proposal for multi-class imbalanced datasets. In ODBOT, the data is first clustered using WBBA-KM clustering to find the dissimilarity relationship between the points of minority classes and majority class. The next step in ODBOT is to detect outliers in minority classes by analysing the difference relationship between the centre of clusters in the minority classes and the majority class. The last step is to generate the samples according to the boundary of the best cluster of the minority class. This solution used datasets of a maximum number of attributes equaling 90 only. So, its performance on datasets with higher dimensionalities is unexamined.

Another proposal is the selective oversampling approach (SOA) (Gnip et al., 2021). In SOA, the outliers in the minor class are first identified using one class support vector machine (OCSVM) then removed. After removing outliers, the data is oversampled to be introduced for the underlying classifier. Gnip et al. used SMOTE and Adaptive Synthetic Sampling Approach (ADASYN) to balance the training data. This proposal used only four real-world datasets with a maximum number of attributes equaling 20 while for the other synthetic datasets 40. Also, the GM score is the only used evaluation metric. So, its performance on real-world datasets with higher dimensionalities or using other metrics is unknown.

Also, SMOTE-IPF (Sáez et al., 2015), is proposed as a combination between oversampling techniques and noise filters. In SMOTE-IPF the training data is first oversampled using SMOTE then the noisy and border line samples are removed using iterative-Partitioning Filter (IPF). This Proposal was evaluated using real-world datasets with a maximum number of attributes equaling 19 only and the minimum imbalance ratio equating (35:301). So, its performance on other datasets with lower imbalance ratios or with higher dimensionalities is unillustrated. Also, it is evaluated using only C4.5 classifier using AUC metric while the performance is unevaluated for other classifiers or metrics.

In (Puri and Kumar Gupta, 2021) a hybrid model is proposed named K-Means-SMOTE-ENN which combines bagging, K-Means clustering, Edited Nearest Neighbours (ENN) and Ad-Boost. It first clusters the training data within each subsample using K-Means clustering. Inside each cluster, the data is oversampled using SMOTE. Then, the noise is removed by using ENN. This Proposal was evaluated using real-world datasets with a maximum number of attributes equaling 9 only. So, its performance on other datasets with higher dimensionalities is unexplained. This is besides its time computational cost ranking the last among all other fifteen approaches compared to it.

Also, Denoise-Based Oversampling (DNOS) is another proposal (Dan and Yian, 2020) in which the noise is removed from the minor and major samples based on K-means clustering. Then the resulting training data is oversampled by using ADASYN. This solution was evaluated using only a single classifier which is SVM using

real-world datasets with minimum imbalance ratios equaling (51:477). So, its performance on datasets with lower imbalance ratios or other classifiers is unknown.

Also, Noise-Adaptive Synthetic Oversampling TECHnique (NASOTECH) (Vo et al., 2021) is another proposal in which Noise-Adaptive Synthetic Oversampling (NASO) rule is used. The NASO utilized the K-Nearest Neighbours for each sample in the minor class to find the total distance from this sample to all of its Kth nearest neighbours. Based on the calculated total distances, they calculated the noise ratio for each sample in the minority class. Next, noise ratio was used to determine the number of samples to be synthesized for each minor sample. This solution was evaluated using only a single classifier which is SVM using real-world datasets with maximum number of features equaling 294. So, its performance on other classifiers with different operation theories is unrecognized.

Those studies prove that integrating SMOTE with outlier cleaning methods will promote the performance of different machine learning classifiers by feeding them with low noise balanced training datasets. However, some of them apply their methods on limited number of classifiers and their performance is unknown on other types of classifiers. Others, do not examine some important evaluation metrics. All the previous works evaluate the performance of their methods on datasets with limited number of attributes. Moreover, all of them do not consider datasets with very critical imbalanced ratios.

These observations have led us to the following research questions:

- **RQ1:** Can RN-SMOTE enhance the performance of different classifier types?
- **RQ2:** Can RN-SMOTE outperform current state of art method?
- **RQ3:** Can RN-SMOTE improve the performance of the classifiers on real datasets with different dimensionalities?
- **RQ4:** Can RN-SMOTE improve the performance of the classifiers on real datasets with critical imbalance ratio?

3. Proposed methodology (RN-SMOTE)

The proposed model **RN-SMOTE** is a combination between SMOTE and DBSCAN to produce high quality oversampled training datasets. In this section, we give a brief introduction to both SMOTE and DBSCAN. Then, **RN-SMOTE** is described in details.

3.1. Synthetic Minority Over-sampling Technique (SMOTE)

Synthetic Minority Over-sampling Technique or as it is abbreviated as SMOTE is a popular and widely used oversampling technique used to solve the imbalanced classification problem (Chawla et al., 2002). In SMOTE, the minority class is oversampled by generating new samples from each sample in the minority class and its nearest neighbours where each synthesised sample X_{syn} is given by:

$$X_{syn} = X_i + \text{rand}(0, 1) \times |X_i - X_{\text{neighbour}}| \quad (1)$$

where X_i is a given sample from the minority class, $X_{\text{neighbour}}$ is a randomly selected sample from the K nearest neighbours to the sample X_i and $\text{rand}(0, 1)$ is a random number between 0 and 1. Fig. 1 shows how synthesised samples are generated from the minority class (Borovicka et al., 2012). SMOTE adds a synthetic instance, at random, along the line joining two real instances. The pseudocode of SMOTE is given in algorithm 1 (Krawczyk et al., 2017).

The main advantage of using SMOTE for oversampling rather than using other methods such as Random Over Sampling (ROS)

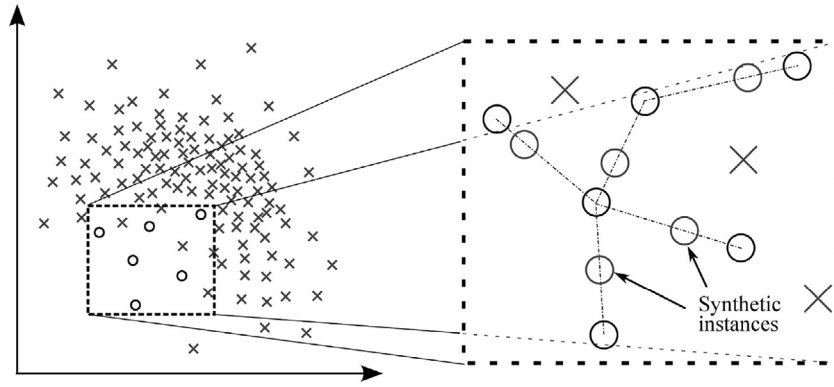


Fig. 1. Generation of Synthetic Instances using SMOTE (Borovicka et al., 2012).

is the avoidance of overfitting. This can be achieved by synthesizing new samples from the minor class instead of replicating them as occurs in ROS (Jeatrakul et al., 2010). Despite the popularity and good performance in different application areas, unfortunately nothing is absolutely perfect, so SMOTE has many problems. The first problem is the high dimensionality problem of datasets being used that limits most classifiers from increasing their performance despite oversampling the minor class (Blagus and Lusa, 2013). Another problem is the negligence of neighbours from the major class resulting in the generation of minor samples leading to more overlapping between classes especially when the minor class is sparse (Maciejewski and Stefanowski, 2011). The most important limitation of SMOTE regardless of the dataset dimensionality is the generation of noisy samples as a result of its inherent randomness which is obvious in Eq. (1). This is beside the noise that may exist in the original dataset and may be used in the generation of new samples leading to more propagation and increase of noise. These noisy samples act as outliers that mislead many classifiers from increasing their performance while datasets are being over-sampled using SMOTE (Cheng et al., 2019; Rivera, 2017). So, it is a good practice to use one of the noise removal techniques when SMOTE is applied. This helps to reduce the effect of noise generated by SMOTE or noise that may exist in the original datasets which improves the performance of classifiers used to classify the datasets.

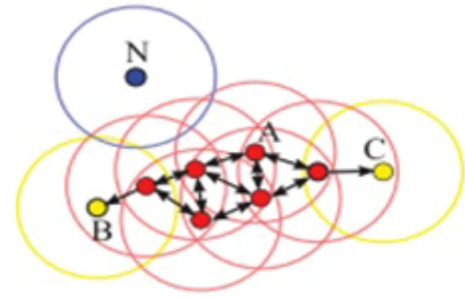


Fig. 2. Basic Concepts of DBSCAN Clustering (Schubert et al., 2017).

3.2. Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

DBSCAN is one of the density-based clustering algorithms family which is used to determine the clusters of any shape that may exist in a given dataset (Ester et al., 1996). DBSCAN uses only two parameters which are the radius of the cluster (*eps*) and the minimum number of points (*MinPts*) within a circle of radius *eps* which is required to create a new cluster (Hao et al., 2015). It also, categorizes all points into three categories which are core points, border points and noise points. Fig. 2 illustrates the basic concepts behind DBSCAN (Schubert et al., 2017). Given a *MinPts* equals 4 and *eps* is the radius of the drawn circles. Then, according to the basic concepts of DBSCAN it is clear that point **N** is categorized as a noise point because it is not reachable from any other point. On the other hand, the point **A** is categorized as a core point, while both **B** and **C** points are categorized as border points because they are density connected and they are reachable from point **A**. The pseudocode of DBSCAN is given in algorithm 2 (Patwary et al., 2012).

One of the main advantages obtained by using DBSCAN as a clustering technique is its ability to recognize the isolated samples which are not reachable from any other points and mark them as

Algorithm 1: SMOTE algorithm

Input: X_{minor} , N_{percent} , K

Output: X_{smote}

Function SMOTE (X_{minor} , N_{percent} , K)

1. $X_{\text{smote}} \leftarrow \{\}$
2. for $i \leftarrow 1$ to $\text{Len}(X_{\text{minor}})$ do
3. $nn \leftarrow K$ Nearest Neighbors (X_i , X_{minor} , k)
4. $p \leftarrow \lfloor N_{\text{percent}}/100 \rfloor$
5. while $p! = 0$ do
6. $X_{\text{neighbour}} \leftarrow \text{select random}(nn)$
7. $X_{\text{smote}} \leftarrow X_i + \text{rand}(0, 1) * |X_{\text{neighbour}} - X_i|$
8. $p \leftarrow p - 1$
9. end while
10. end for
11. return X_{smote}



Fig. 3. Overview of the RN- SMOTE Mode showing the steps and both of minor and major class with each step.

noise (Bushra and Yi, 2021; Huang et al., 2017). So, according to Patel and Thakral (2016), DBSCAN is considered as one of the best clustering algorithms that can be used in data mining tasks to detect outliers. Another advantage of DBSCAN is the auto-discovery of the clusters that exist in a given dataset and hence it does not require the number of clusters to be externally determined by the user (Zhou et al., 2015).

An important practice while applying DBSCAN is the determination of its parameters which are **eps** and **MinPts**. The Sorted K-distance graph was the first and still widely used technique for DBSCAN parameters estimation (Elbatta et al., 2013; Starczewski et al., 2020; Wang et al., 2015). In this technique, the Kth nearest neighbours for each point are found then sorted in ascending order according to the distances of the Kth nearest neighbours. Then, we find the maximum curvature point for the resulting curve from which we find the value of **eps**.

Algorithm 2: DBSCAN

Input: **eps**, **minpts**, **X**

Output: set of clusters

Procedure DBSCAN (**X**, **eps**, **minpts**)

```

1. for each point  $P \in X$  do
2.   if label ( $P$ )  $\neq$  visited then
3.     label ( $P$ )  $\leftarrow$  visited
4.    $N \leftarrow$  Neighbors ( $P$ , eps)
5.   if  $N < \text{minpts}$  then
6.     mark  $P$  as Noise
7.   else
8.      $C \leftarrow \{P\}$ 
9.     for each point  $P_n \in N$  do
10.       $N \leftarrow N \setminus P_n$ 
11.      if label ( $P_n$ )  $\neq$  visited then
12.        label ( $P_n$ )  $\leftarrow$  visited
13.         $N_n \leftarrow$  Neighbors ( $P_n$ , eps)
14.        if  $N_n \geq \text{minpts}$  then
15.           $N \leftarrow N \cup N_n$ 
16.        if  $P_n$  not a member of any cluster then
17.           $C \leftarrow C \cup P_n$ 
18.      end if
19.    end if
20.  end if
21. end for
22. end if
23. end if
24. end for

```

3.3. RN- Smote

In this section, we describe the proposed Reduced Noise-SMOTE. **RN-SMOTE** combines DBSCAN with SMOTE oversampling

in order to obtain low noise training data before applying the learning algorithm. The basic idea is to apply SMOTE on the training data which is unbalanced. The result of SMOTE is balanced training data that contains noise that may mislead the classifier that will be fitted on the data. So, in the next stage, DBSCAN is applied and the detected noise is removed which results in clean but unbalanced data which is combined with the training data. To balance this data to be ready for a classifier, SMOTE is applied again resulting in clean balanced data. Fig. 3 shows an overview of this process.

To determine the **eps** value for DBSCAN, the K-distance graph technique is used. For each sample in the training set, the Kth nearest neighbours are determined and then sorted in ascending order according to the calculated distance. A K-distance graph is then plotted. The maximum curvature point is determined from which **eps** is determined. Fig. 4 plots the K-distance graph for an example of one of the datasets used in this paper which is the cardiocography dataset with marked maximum curvature point and optimal **eps** value.

To find the **MinPts** value, we have assumed a value **K** parameter to be used for finding the Kth nearest neighbours and also used as the minimum number of points required to start a new cluster (**MinPts**). The assumed value of **K** is chosen to be proportional to the number of samples in the training set. We have tried many values of **K** and estimated the performance through many metrics. The finally chosen **K** value is the natural log of the number of samples in the training set which is given by:

$$K = \ln(N) \quad (2)$$

where **N** is the number of samples in the used training set. The steps to find the **eps** with assumed value of **K** is given in pseudocode in algorithm 3.

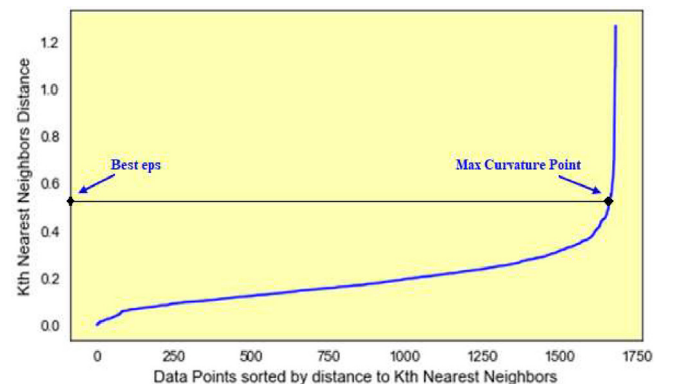


Fig. 4. K-distance Graph for Cardiocography dataset with marked optimal **eps** value.

Algorithm 3: FindEpsInput: X_{minor} Output: ϵps **Function FindEps (X_{minor})**

```

1.  $N \leftarrow \text{length}(X_{minor})$ 
2.  $K \leftarrow \text{Ln}(N)$ 
3. for  $i \leftarrow 1$  to  $N$ 
4.    $nn \leftarrow \text{Nearest Neighbors}(X_i, k)$ 
5.   for  $j \leftarrow 1$  to  $\text{length}(nn)$ 
6.      $D(X_i, X_j) \leftarrow \text{find distance}(X_i, X_j)$ 
7.   end for
8.    $\text{Distances} \leftarrow \text{MIN } D(X_i, X_j)$ 
9. end for
10.  $\text{Distances} \leftarrow \text{Sort distances ascending}$ 
11.  $\text{Knee} \leftarrow \text{KneeLocator}(\text{Distances})$ 
12.  $\epsilon ps \leftarrow \text{Knee}(\text{knee\_y})$ 
13. return  $\epsilon ps$ 

```

Fig. 5 shows the details of the proposed **RN-SMOTE**. As shown in the steps in Fig. 5, the whole process is started with splitting each used dataset using 10-fold cross validation. One-fold is reserved for the model evaluation while the other 9 folds are used for training the model. The SMOTE is applied to the training sets only resulting in oversampled balanced but noisy training set. The minor class samples are extracted to be handled by DBSCAN. Using the K value in Eq. (2), the K th nearest neighbours are determined and the K -distances are computed then the **eps** value is calculated using pseudocode in algorithm 3.

The calculated **eps** and K -value (**MinPts**) are passed as parameters to DBSCAN, to cluster the minor samples and generates the noise cluster. The samples in the resulted noise cluster are removed. The resulted clean minors are then combined with the original training set. Each used classifier is trained on this combination while the previously reserved test fold is used now to evaluate each trained classifier.

4. Experiments and results discussion

In this section, we discuss the evaluation of RN-SMOTE. This discussion includes the performance evaluation of 9 different classification algorithms. This evaluation has been carried out using nine different real-world datasets with different imbalance ratios (ranging from 1.94% to 50.92%) and different dimensionalities (number of features ranges from 5 to 1557) using six different evaluation metrics used widely in the evaluation of classifier's performance in machine learning and data mining. Our experiments are organized to answer the research questions which have introduced in Section 2.

4.1. Datasets

This section provides brief characteristics of the acquired datasets used in this paper. In the experiments that have been carried out to evaluate the proposed model in this paper, 9 different imbalanced datasets from different application areas and representing binary classification problems are used. These datasets have a different number of features and different imbalance ratios. Table 1 lists each one of the 9 datasets each with its characteristics and

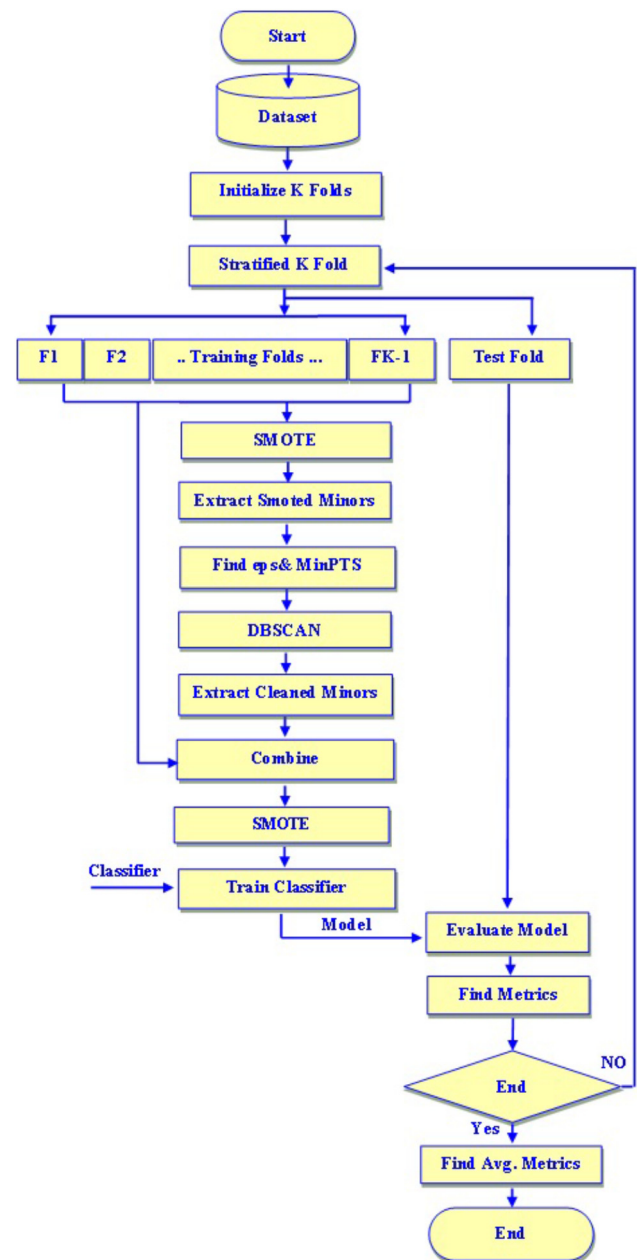


Fig. 5. The Flowchart of RN-SMOTE.

all datasets are sorted according to their imbalance ratios. The first 5 datasets in the table are all normalized so, in our experiments none of them will be normalized. These datasets were prepared to be used for outlier detection research (Campos et al., 2016) and downloaded from the repository in (<https://www.dbs.ifi.lmu.de/research/outlier-evaluation/DAMI/>, n.d.). These datasets have been chosen as they are already used for outlier detection and also, they are imbalanced that is suitable for the purpose of this paper. The other 4 datasets are not normalized, so in our experiments, they are normalized using Standard Scaler. They were used by (Zhang et al., 2021) for imbalanced classification while downloaded from (<https://archive.ics.uci.edu/ml/datasets/ILPD>, n.d.; Lucas et al., 2013; Mansouri et al., 2013; Yeh et al., 2009).

Table 1

Datasets Description.

Dataset	#Features	#Sample	#Minors	#Majors	Normalized?	Imbalance Ratio
Stamps	11	315	6	309	Yes	1.94%
Internet Ads	1557	1630	32	1598	Yes	2%
Cardiotocography	23	1681	33	1648	Yes	2%
WBC	11	454	10	444	Yes	2.25%
Lymphography	20	148	6	142	Yes	4.225%
Climate	21	540	46	494	No	9.31%
Blood	5	748	178	570	No	31.22%
ILPD	11	583	167	416	No	40.14%
QSAR	42	1055	356	699	No	50.92%

4.2. Evaluation Metrics

In machine learning classification tasks, confusion matrix parameters which are True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) are the main parameters from which other performance metrics such as Precision, Recall and F1 score are computed (Martin Ward Powers, 2011). Matthew's Correlation Coefficient (MCC) is another metric that is used in evaluating classification models with imbalanced datasets to indicate how much the predictive model is better than random guess (Jiao and Du, 2016). Cohesion's Kappa is a metric that measures the agreement between predicted and true classes (Grandini et al., 2020). Another important metric is the Geometric-Mean (GM) score which results from the product of sensitivity and specificity and it is considered the most reliable performance metric for imbalanced classification (Helal et al., 2016). An important advantage of GM score is measuring how overfitting neglect both minor and major classes, so it is mainly used for evaluating classifiers with imbalanced datasets (Wang et al., 2021). Table 2 lists all metrics used for evaluating the classifiers used in this paper with their expressions as calculated based on confusion matrix parameters.

4.3. Experimental Results Discussion

Resampling techniques have a notable effect on the performance of the machine learning classifiers. Based on the nature of the used dataset and the used classifier, the performance of many classifiers often increases while the dataset is being oversampled using any of the oversampling techniques such as SMOTE, Random Over Sampling (ROS), etc.

In the experiments that have been carried out in this paper, 9 classifiers have been utilized. These classifiers are Classification and Regression Tree (CART), Gradient Boost (GB), Naïve Bayes (NB), Random Forest (RF), K-Nearest Neighbours (KNN), eXtreme Gradient Boosting (XGBOOST), Logistic Regression with Stochastic Gradient Descent- (SGD-LR), Support Vector Machines (SVM) and Adaptive Boosting (Ada-Boost). These classifiers have been used as result of their popularity in machine learning with different application areas. All of these classifiers have been used by their default settings in python libraries and none of their hyperparameters have been optimized.

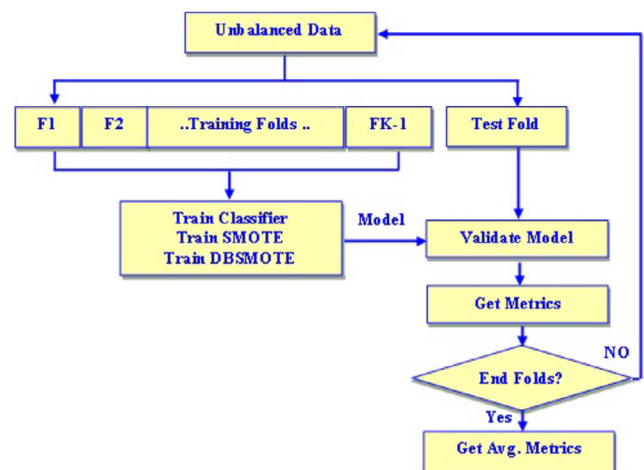
To evaluate the proposed **RN-SMOTE** model, we have compared the performance of each of the previously mentioned classifiers in

three cases. The first case is training the classifier using original data only without any sampling. The second case is training the classifier using the original data after oversampling using SMOTE. Finally, the third case is training the classifier with the original data after oversampling using RN-SMOTE. The performance has been measured using all metrics discussed in the previous section and listed in Table 2. For a fair evaluation, in each split of the 10-fold cross validation, all classifiers in the previously mentioned three cases are trained on the same training folds and evaluated using the same test fold as shown in Fig. 6. Also, to accurately measure the performance of each classifier in the three cases, we have repeated the training and evaluation experiment 100 times. In each of the 100 times, each classifier is trained and evaluated using 10-fold stratified cross validation and the average metrics are computed on these 10 folds. These average metrics are then grouped for the 100 times and then their average is computed to obtain the final average metrics in each of the three cases.

4.4. Results Discussion for All Classifiers using the Blood Dataset

In this section we will discuss the results of all classifiers on the Blood dataset. Blood dataset has 748 samples, the major class has 570 samples while the minor class has 178 samples with a 31.22% imbalanced ratio. Fig. 7 draws a comparison among the performance of all of the used classifiers in the previously mentioned three test cases in terms of the GM score using the Blood dataset. Fig. 8 draws the comparison in terms of the minor Recall score.

For the **CART classifier**, the performance based on GM score has increased by 2.55% for **RN-SMOTE** model when compared with SMOTE and by 3.9% when compared with the original dataset.

**Fig. 6.** Training and evaluation of classifiers in RN-SMOTE.**Table 2**

Imbalanced Classification Metrics.

Metric	Expression	Metric	Expression
GM	$\sqrt{\frac{TP}{TP+FN} \times \frac{TN}{TN+FP}}$	Precision	$\frac{TP}{TP+FP}$
MCC	$\frac{TP-TN-FP-FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	Recall	$\frac{TP}{TP+FN}$
Kappa	$\frac{Po-Pe}{1-Pe}$	F1	$\frac{2 \times \text{precision} \times \text{Recall}}{\text{precision} + \text{Recall}}$

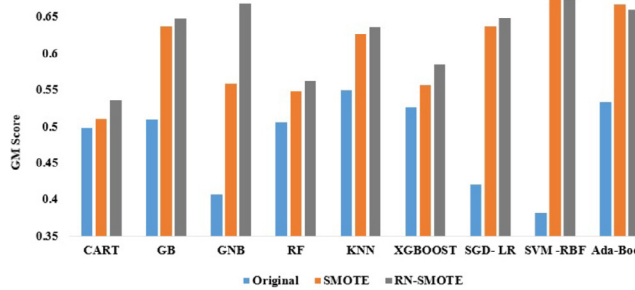


Fig. 7. all classifiers GM score for Blood dataset.

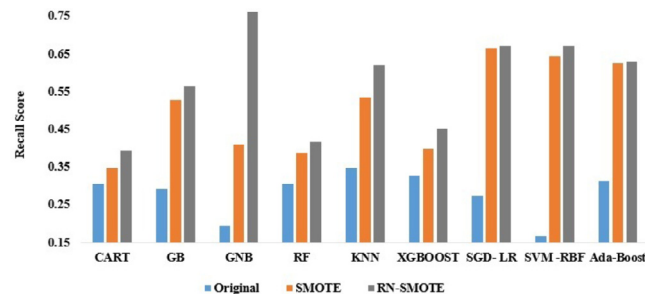


Fig. 8. All classifiers Recall scores for Blood dataset.

While, the performance based on minor Recall score has increased by 4.65% for the proposed model when compared with SMOTE and by 8.73% when compared with the original dataset.

For the GB classifier, the performance based on GM score has increased by 1% for **RN-SMOTE** model when compared with SMOTE and by 13.79% when compared with the original dataset. While the performance based on minor Recall score has increased by 3.67% for the proposed model when compared with SMOTE and by 27.22% when compared with the original dataset.

For the NB classifier, the performance based on GM score has increased by 10.94% for **RN-SMOTE** model when compared with SMOTE and by 26.16% when compared with the original dataset. While the performance based on minor Recall score has increased by 35.17% for the proposed model when compared with SMOTE and by 56.88% when compared with the original dataset.

Table 3
Results summary for CART classifier.

Classifier: Classification and Regression Tree (CART)							
Dataset	Metrics						
	Model	GM	Kappa	MCC	F1	Precision	Recall
Cardiotocography	Original	0.6342	0.4982	0.5145	0.5057	0.5935	0.4863
	SMOTE	0.7729	0.6158	0.6305	0.6227	0.6578	0.6476
	RN-SMOTE	0.7917	0.6479	0.6632	0.6541	0.6982	0.6723
Blood	Original	0.4976	0.1567	0.1598	0.3335	0.3793	0.3051
	SMOTE	0.5111	0.124	0.1254	0.3345	0.332	0.3459
	RN-SMOTE	0.5366	0.1382	0.1403	0.3575	0.3348	0.3924

The bold value means the best (maximum) value among every 3 values in the column of each metric with each dataset as a row.

Table 4
Results summary for GB classifier.

Classifier: Gradient Boost (GB)							
Dataset	Metrics						
	Model	GM	Kappa	MCC	F1	Precision	Recall
Stamps	Original	0.441	0.2213	0.1749	0.1757	0.1601	0.211
	SMOTE	0.6074	0.3969	0.2984	0.2961	0.2735	0.345
	RN-SMOTE	0.5964	0.3947	0.3027	0.3006	0.2781	0.349
Blood	Original	0.5095	0.2509	0.2723	0.3731	0.5467	0.2922
	SMOTE	0.6374	0.2885	0.2937	0.473	0.4365	0.5277
	RN-SMOTE	0.6474	0.2838	0.2921	0.4783	0.4208	0.5644
ILPD	Original	0.495	0.1602	0.1675	0.3521	0.443	0.3013
	SMOTE	0.6118	0.235	0.2396	0.473	0.4364	0.5261
	RN-SMOTE	0.6219	0.2427	0.2492	0.4845	0.4356	0.5555
QSAR	Original	0.8348	0.6928	0.6963	0.7914	0.8306	0.76
	SMOTE	0.8487	0.7	0.7021	0.8014	0.8011	0.8059
	RN-SMOTE	0.8488	0.6971	0.6993	0.8003	0.794	0.8111

The bold value means the best (maximum) value among every 3 values in the column of each metric with each dataset as a row.

For the RF classifier, the performance based on GM score has increased by 1.47% for **RN-SMOTE** model when compared with SMOTE and by 5.66% when compared with the original dataset. While the performance using minor Recall score has increased by 3.04% for the proposed model when compared with SMOTE and by 11.18% when compared with the original dataset.

For the KNN classifier, the performance based on of the GM score has increased by 0.87% for **RN-SMOTE** model when compared with SMOTE and by 8.62% when compared with the original dataset. While the performance based on minor Recall score has increased by 8.67% for the proposed model when compared with SMOTE and by 27.39% when compared with the original dataset.

For the XGBOOST classifier, the performance based on GM score has increased by 2.8% for **RN-SMOTE** model when compared with SMOTE and by 5.83% when compared with the original dataset. While the performance based on minor Recall score has increased by 5.35% for the proposed model when compared with SMOTE and by 12.6% when compared with the original dataset.

For the SGD-LR classifier, the performance based on GM score has increased by 1.09% for **RN-SMOTE** model when compared with SMOTE and by 22.78% when compared with the original dataset. While the performance based on minor Recall score has increased by 0.71% for the proposed model when compared with SMOTE and by 39.85% when compared with the original dataset.

Table 5
Results summary for NB classifier.

Classifier: Naïve Bayes (NB)							
Dataset	Metrics						
	Model	GM	Kappa	MCC	F1	Precision	Recall
Stamps	Original	0.4822	0.2495	0.1808	0.1842	0.1765	0.2
	SMOTE	0.5081	0.2603	0.1826	0.1846	0.1761	0.202
	RN-SMOTE	0.497	0.2655	0.1926	0.1943	0.185	0.214
Cardiotocography	Original	0.5848	0.2408	0.2659	0.2609	0.2073	0.4404
	SMOTE	0.7934	0.2258	0.3054	0.2507	0.1578	0.7408
	RN-SMOTE	0.7988	0.1935	0.2817	0.2202	0.134	0.7618
Climate	Original	0.1154	0.0136	0.0222	0.1667	0.0982	0.9707
	SMOTE	0.3671	0.0425	0.1151	0.1898	0.1085	0.9752
	RN-SMOTE	0.418	0.0545	0.1356	0.1993	0.1149	0.9749
Blood	Original	0.407	0.1452	0.1661	0.2644	0.4526	0.193
	SMOTE	0.5592	0.2208	0.2288	0.398	0.4168	0.4101
	RN-SMOTE	0.6686	0.2621	0.3061	0.4975	0.3724	0.7618
ILPD	Original	0.6136	0.2472	0.3492	0.5534	0.3906	0.9532
	SMOTE	0.6263	0.2582	0.3548	0.5576	0.3967	0.9429
	RN-SMOTE	0.6489	0.2810	0.3691	0.5672	0.4093	0.929
QSAR	Original	0.7192	0.4135	0.4869	0.6706	0.5182	0.9531
	SMOTE	0.7404	0.4448	0.5082	0.684	0.5377	0.9431
	RN-SMOTE	0.7531	0.4645	0.5202	0.6922	0.5516	0.9325

The bold value means the best (maximum) value among every 3 values in the column of each metric with each dataset as a row.

Table 6
Results summary for RF classifier.

Classifier: Random Forest (RF)							
Dataset	Metrics						
	Model	GM	Kappa	MCC	F1	Precision	Recall
InternetADS	Original	0.6166	0.562	0.5997	0.5658	0.8349	0.4557
	SMOTE	0.661	0.6065	0.6413	0.6102	0.8632	0.5039
	RN-SMOTE	0.6655	0.6109	0.6439	0.6146	0.8563	0.5119
WBC	Original	0.4872	0.465	0.4678	0.4667	0.4565	0.488
	SMOTE	0.4982	0.3789	0.3956	0.3884	0.3426	0.503
	RN-SMOTE	0.5291	0.4098	0.4266	0.419	0.3733	0.534
Lymphography	Original	0.7129	0.5208	0.3137	0.3137	0.313	0.315
	SMOTE	0.6836	0.4718	0.287	0.287	0.285	0.291
	RN-SMOTE	0.7217	0.5326	0.3211	0.321	0.3195	0.324
Blood	Original	0.5065	0.1975	0.2048	0.3525	0.4359	0.3043
	SMOTE	0.5484	0.1857	0.1878	0.3787	0.3817	0.3857
	RN-SMOTE	0.5631	0.1912	0.1937	0.3911	0.3774	0.4161
QSAR	Original	0.8377	0.7037	0.7081	0.7975	0.8472	0.7576
	SMOTE	0.8484	0.7122	0.7148	0.8063	0.8313	0.7868
	RN-SMOTE	0.8478	0.7137	0.7167	0.8067	0.8369	0.7829

The bold value means the best (maximum) value among every 3 values in the column of each metric with each dataset as a row.

For the SVM classifier, the performance based on GM score has a degradation by 0.11% for **RN-SMOTE** model when compared with SMOTE but it has an increase by 29.29% when compared with the original dataset. While the performance based on minor Recall score has increased by 2.76% for the proposed model when compared with SMOTE and by 50.49% when compared with the original dataset.

For the Ada-Boost classifier, the performance based on GM score has a degradation by 0.65% for **RN-SMOTE** model when compared with SMOTE but it has an increase by 12.73% when compared with the original dataset. While the performance based on minor Recall score has increased by 0.32% for the proposed model when compared with SMOTE and by 31.48% when compared with the original dataset.

This section is the answer for RQ1. It is clear that the proposed **RN-SMOTE** model promotes the performance of different machine learning classifiers by feeding them with low noise balanced training datasets. On Blood dataset, the best classifier that has gained an improvement with our proposed model is NB classifier.

4.5. Results Discussion for Classifiers

In this section, the performance of each classifier will be discussed. The GM, Kappa, MCC, minor class F1, minor class Precision and minor class Recall scores have been measured for each classifier with each of the nine datasets in each of the three evaluation cases. The results for each classifier are listed in a separate table containing only results for datasets with which the classifiers combined with **RN-SMOTE** have recorded an increase in their performance in terms of any of the mentioned metrics compared to the other two cases. The datasets with which the classifiers combined with our proposal had a lower performance in all metrics compared to any of the other cases are not listed. The best results are bolded in the tables.

The results for the CART classifier with all datasets that resulted in an increase in the CART performance in terms of any metric compared to the other two cases are listed in Table 3. When trained and evaluated using Cardiotocography dataset, the results show that the performance of CART combined with our

Table 7
Results summary for KNN classifier.

Classifier: K-Nearest Neighbors (KNN)							
Dataset	Metrics						
	Model	GM	Kappa	MCC	F1	Precision	Recall
Stamps	Original	0.399	0	0	0	0	0
	SMOTE	0.6161	0.4312	0.3564	0.3517	0.3201	0.423
	RN-SMOTE	0.6263	0.4355	0.3514	0.3472	0.3169	0.415
WBC	Original	0.0079	0.0037	0.0039	0.0057	0.0048	0.008
	SMOTE	0.79	0.5617	0.5957	0.5729	0.4862	0.8
	RN-SMOTE	0.7815	0.5638	0.596	0.5745	0.4908	0.791
Lymphography	Original	0.4	0	0	0	0	0
	SMOTE	0.8334	0.6993	0.4684	0.4675	0.4517	0.5
	RN-SMOTE	0.8341	0.7077	0.468	0.467	0.4542	0.493
Climate	Original	0.3361	0.2518	0.3009	0.2694	0.5691	0.1854
	SMOTE	0.7627	0.281	0.3462	0.3729	0.2504	0.7607
	RN-SMOTE	0.7568	0.3057	0.3594	0.3914	0.2735	0.7235
Blood	Original	0.55	0.2732	0.2857	0.4093	0.5244	0.3461
	SMOTE	0.6275	0.2531	0.2598	0.4551	0.4017	0.5333
	RN-SMOTE	0.6362	0.2274	0.2462	0.4579	0.3662	0.62
ILPD	Original	0.5098	0.1393	0.142	0.3614	0.4022	0.3366
	SMOTE	0.6069	0.1853	0.2053	0.48	0.3807	0.6565
	RN-SMOTE	0.6105	0.1964	0.211	0.4769	0.3918	0.6173

The bold value means the best (maximum) value among every 3 values in the column of each metric with each dataset as a row.

Table 8
Results summary for XGBOOST classifier.

Classifier: eXtreme Gradient Boosting (XGBOOST)							
Dataset	Metrics						
	Model	GM	Kappa	MCC	F1	Precision	Recall
Stamps	Original	0.4994	0.279	0.2001	0.2012	0.1962	0.212
	SMOTE	0.5731	0.3638	0.2832	0.2808	0.2569	0.334
	RN-SMOTE	0.5829	0.3735	0.2914	0.2885	0.263	0.345
InternetADS	Original	0.7051	0.6403	0.6671	0.6445	0.8428	0.5579
	SMOTE	0.7496	0.6461	0.6655	0.6513	0.7719	0.6112
	RN-SMOTE	0.7591	0.6466	0.6649	0.652	0.7541	0.6257
Cardiotocography	Original	0.8107	0.7485	0.7662	0.7521	0.8832	0.694
	SMOTE	0.8496	0.7526	0.7654	0.7569	0.8158	0.7498
	RN-SMOTE	0.8521	0.7566	0.7696	0.7608	0.8199	0.7541
Climate	Original	0.6609	0.5506	0.5818	0.5751	0.7918	0.4865
	SMOTE	0.7981	0.619	0.6313	0.6511	0.6617	0.686
	RN-SMOTE	0.7884	0.6258	0.6383	0.6558	0.6916	0.6668
Lymphography	Original	0.567	0.2777	0.1666	0.167	0.167	0.167
	SMOTE	0.6819	0.4807	0.2972	0.2987	0.298	0.3
	RN-SMOTE	0.7524	0.5953	0.374	0.3747	0.372	0.38
Blood	Original	0.5268	0.2255	0.2334	0.3761	0.4623	0.3259
	SMOTE	0.5571	0.196	0.1982	0.3878	0.3876	0.3984
	RN-SMOTE	0.5851	0.2148	0.2181	0.4136	0.3891	0.4519
ILPD	Original	0.5521	0.2157	0.2203	0.4138	0.4694	0.3797
	SMOTE	0.6079	0.2604	0.2633	0.4727	0.4732	0.4829
	RN-SMOTE	0.6085	0.2639	0.2669	0.4739	0.4769	0.4815

The bold value means the best (maximum) value among every 3 values in the column of each metric with each dataset as a row.

RN-SMOTE outperformed the performance with each of the original datasets and the SMOTE in all metrics. With the blood dataset, the CART combined with our proposal had a hybrid performance and outperformed the other two cases in GM, minor F1 and minor Recall scores.

The results for the GB classifier with all datasets that resulted in an increase in its performance in terms of any metric are listed in Table 4. The results show the hybrid performance of GB classifier combined with our **RN-SMOTE** when compared to the other two cases. When trained and evaluated using ILPD dataset, the performance of GB combined with our model outperformed the other

two cases in all metrics except minor Precision score. While with Stamps dataset, it outperformed all other cases in all metrics except GM and Kappa. Also with Blood dataset, it outperformed only in GM, minor F1 and minor Recall scores. Finally, with QSAR dataset it outperformed only in GM and minor Recall scores.

The results for the NB classifier with all datasets that resulted in an increase in its performance in terms of any metric are listed in Table 5. When trained and evaluated using Climate dataset, the results show that, the performance of the NB classifier combined with our **RN-SMOTE** outperformed the performance with each of the original dataset and the SMOTE in all metrics except Recall.

Table 9
Result summary for SGD-LR classifier.

Classifier: Logistic Regression with Stochastic Gradient Descent- (SGD- LR)							
Dataset	Metrics						
	Model	GM	Kappa	MCC	F1	Precision	Recall
InternetADS	Original	0.6933	0.6387	0.6698	0.6424	0.8713	0.5438
	SMOTE	0.7377	0.6762	0.7023	0.68	0.8732	0.5959
	RN-SMOTE	0.7402	0.677	0.7038	0.6809	0.8772	0.5968
Cardiotocography	Original	0.4623	0.401	0.4394	0.4047	0.6556	0.3177
	SMOTE	0.8892	0.3347	0.4199	0.3553	0.2323	0.8618
	RN-SMOTE	0.895	0.3367	0.4235	0.3573	0.2331	0.8731
WBC	Original	0.3862	0.3149	0.3246	0.3211	0.2929	0.389
	SMOTE	0.7799	0.5686	0.5997	0.5786	0.4957	0.789
	RN-SMOTE	0.789	0.5797	0.6105	0.5894	0.5074	0.798
Lymphography	Original	0.7491	0.5787	0.3583	0.3583	0.354	0.367
	SMOTE	0.7551	0.5941	0.3718	0.372	0.368	0.38
	RN-SMOTE	0.7602	0.6009	0.3742	0.3743	0.3705	0.382
Blood	Original	0.4211	0.1674	0.1867	0.292	0.4194	0.2727
	SMOTE	0.638	0.2491	0.2785	0.4727	0.3839	0.6641
	RN-SMOTE	0.6489	0.2596	0.2889	0.4806	0.388	0.6712
ILPD	Original	0.506	0.1798	0.1902	0.3739	0.4329	0.3764
	SMOTE	0.632	0.2524	0.2813	0.5091	0.4236	0.6755
	RN-SMOTE	0.6428	0.2663	0.2925	0.5185	0.4325	0.6785
QSAR	Original	0.8129	0.6375	0.6416	0.7577	0.7715	0.7532
	SMOTE	0.8338	0.6511	0.657	0.7756	0.7375	0.8249
	RN-SMOTE	0.8409	0.6687	0.6733	0.7856	0.7546	0.8254

The bold value means the best (maximum) value among every 3 values in the column of each metric with each dataset as a row.

Table 10
Result summary for SVM classifier.

Classifier: Support Vector Machines (SVM)							
Dataset	Metrics						
	Model	GM	Kappa	MCC	F1	Precision	Recall
Stamps	Original	0.4	0	0	0	0	0
	SMOTE	0.5706	0.3829	0.3271	0.5806	0.387	0.4
	RN-SMOTE	0.5806	0.387	0.3267	0.3219	0.2883	0.4
InternetADS	Original	0.5017	0.4531	0.4978	0.4564	0.7609	0.3418
	SMOTE	0.5814	0.5272	0.5661	0.531	0.8055	0.4208
	RN-SMOTE	0.582	0.5279	0.5667	0.5317	0.8057	0.4216
Cardiotocography	Original	0.2428	0.2056	0.2412	0.2077	0.43	0.1393
	SMOTE	0.8183	0.4617	0.4961	0.4754	0.3738	0.7186
	RN-SMOTE	0.8349	0.4852	0.5198	0.4983	0.3953	0.7432
WBC	Original	0	0	0	0	0	0
	SMOTE	0.5491	0.4049	0.4252	0.4153	0.3593	0.555
	RN-SMOTE	0.5639	0.4159	0.4368	0.426	0.3682	0.57
Lymphography	Original	0.4	0	0	0	0	0
	SMOTE	0.602	0.3367	0.202	0.202	0.202	0.202
	RN-SMOTE	0.617	0.3617	0.217	0.217	0.217	0.217
Climate	Original	0.2876	0.2229	0.2779	0.2344	0.5559	0.1533
	SMOTE	0.7026	0.5598	0.5785	0.5893	0.7094	0.5436
	RN-SMOTE	0.6983	0.5737	0.5978	0.6002	0.7631	0.535
QSAR	Original	0.8523	0.7237	0.7264	0.8132	0.8458	0.7866
	SMOTE	0.8677	0.7274	0.7299	0.8219	0.8026	0.8457
	RN-SMOTE	0.8684	0.7313	0.7335	0.8238	0.8094	0.8424

The bold value means the best (maximum) value among every 3 values in the column of each metric with each dataset as a row.

With Stamps, Blood, ILPD and QSAR it outperformed the other two cases in all metrics except minor Recall for QSAR and ILPD datasets, minor Precision for Blood dataset and GM for Stamps dataset. Also, for the Cardiotocography dataset, it outperformed only in GM and minor Recall scores.

The results for the RF classifier with all datasets that resulted in an increase in its performance in terms of any metric are listed in Table 6. When trained and evaluated using Lymphography dataset, the results show that, the performance of the RF classifier

combined with our **RN-SMOTE** outperformed the performance with each of the original datasets and the SMOTE. With all other listed datasets, the RF classifier had a hybrid performance. For InternetADS dataset, it outperformed the two other cases in all metrics except minor Precision. With QSAR dataset, it outperformed in all metrics except GM, minor Precision and minor Recall. With Blood dataset, it outperformed in GM, minor F1 and Recall. Also, with WBC dataset, it outperformed in GM and minor Recall.

Table 11
Result summary for Ada-Boost classifier.

Classifier: Adaptive Boosting (Ada-Boost)							
Dataset	Metrics						
	Model	GM	Kappa	MCC	F1	Precision	Recall
Stamps	Original	0.5339	0.2891	0.2011	0.2012	0.1922	0.221
	SMOTE	0.5205	0.2848	0.2076	0.2078	0.194	0.237
	RN-SMOTE	0.5396	0.2941	0.2074	0.2077	0.1952	0.234
InternetADS	Original	0.7155	0.6195	0.642	0.6246	0.7702	0.5727
	SMOTE	0.7185	0.6265	0.6497	0.6315	0.7841	0.5758
	RN-SMOTE	0.7295	0.6318	0.6541	0.6369	0.7793	0.5876
Cardiotocography	Original	0.6826	0.604	0.6303	0.6087	0.7943	0.5335
	SMOTE	0.8085	0.6693	0.6832	0.6753	0.7146	0.6927
	RN-SMOTE	0.8164	0.6854	0.7	0.691	0.7396	0.7017
WBC	Original	0.4729	0.4166	0.4239	0.4209	0.3966	0.475
	SMOTE	0.489	0.3836	0.3978	0.3924	0.3498	0.493
	RN-SMOTE	0.5123	0.4162	0.4292	0.424	0.3852	0.516
Lymphography	Original	0.6423	0.4314	0.2933	0.2932	0.2815	0.317
	SMOTE	0.6412	0.4478	0.3152	0.3152	0.303	0.34
	RN-SMOTE	0.6429	0.4478	0.3161	0.3155	0.3022	0.343
Climate	Original	0.6575	0.5	0.5189	0.5329	0.6512	0.4919
	SMOTE	0.7536	0.4618	0.4804	0.5151	0.4547	0.6441
	RN-SMOTE	0.7671	0.5193	0.535	0.564	0.5288	0.6561
ILPD	Original	0.5506	0.2188	0.2244	0.4133	0.4767	0.3753
	SMOTE	0.6417	0.2666	0.2762	0.5077	0.444	0.6025
	RN-SMOTE	0.6459	0.2704	0.2817	0.5128	0.4432	0.6182
QSAR	Original	0.8325	0.6755	0.6776	0.7832	0.7967	0.7743
	SMOTE	0.8389	0.6699	0.6727	0.7847	0.7636	0.8112
	RN-SMOTE	0.8413	0.672	0.6751	0.7868	0.761	0.8185

The bold value means the best (maximum) value among every 3 values in the column of each metric with each dataset as a row.

Table 12
The improved classifiers using RN-SMOTE on different datasets.

Dataset	#Features	Imbalance Ratio	CART	GB	NB	RN-SMOTE + Classifier					
						Classifiers					
						RF	KNN	XG-Boost	SGD-LR	SVM	Ada-Boost
Stamps	11	1.94%	–	✓	✓	–	✓	✓	–	✓	✓
Internet Ads	1557	2%	–	–	–	✓	–	✓	✓	✓	✓
Cardiotocography	23	2%	✓	–	✓	–	–	✓	✓	–	✓
WBC	11	2.25%	–	–	–	✓	✓	–	✓	✓	✓
Lymphography	20	4.225%	–	–	–	✓	✓	✓	✓	✓	✓
Climate	21	9.31%	–	–	✓	–	✓	✓	–	✓	✓
Blood	5	31.22%	✓	✓	✓	✓	✓	✓	✓	–	–
ILPD	11	40.14%	–	✓	✓	–	✓	✓	✓	–	✓
QSAR	42	50.92%	–	✓	✓	✓	–	–	✓	✓	✓

The results for the KNN classifier with all datasets that resulted in an increase in its performance in terms of any metric are listed in Table 7. With all listed datasets, the KNN classifier had a hybrid performance. With WBC dataset, the performance of KNN combined with our **RN-SMOTE** outperformed the other two cases in all metrics except GM and minor Recall. With Stamps dataset, it outperformed only in both GM and Kappa. With Lymphography dataset, it outperformed in GM, Kappa and minor Precision. With Climate dataset, it outperformed in Kappa, MCC and minor F1. With ILPD it outperformed in GM, Kappa and MCC. Finally, with Blood dataset, it outperformed in GM, minor F1 and Recall.

The results for the XGBOOST classifier with all datasets that resulted in an increase in its performance in terms of any metric are listed in Table 8. When trained and evaluated using Stamps and Lymphography datasets the results show that, the performance of the XGBOOST classifier combined with our **RN-SMOTE** outperformed the performance with the two other cases. Also with all other listed datasets, the XGBOOST had a hybrid performance. With Cardiotocography, it outperformed in all metrics except

minor Precision. With ILPD dataset, it outperformed the other two cases in all metrics except minor Recall. With InternetADS dataset, it outperformed in all metrics except MCC and minor Precision. With Climate dataset, it outperformed in Kappa, MCC, and minor F1. Finally, with Blood dataset, it outperformed in GM and minor F1 and Recall.

The results for the SGD-LR classifier with all datasets that resulted in an increase in its performance using any metric are listed in Table 9. When trained and evaluated using InternetADS, WBC, Lymphography and Blood datasets the results show that, the performance of the SGD-LR classifier combined with our **RN-SMOTE** outperformed the performance with each of the original datasets and the SMOTE. With all other listed datasets, it had a hybrid performance. For both ILPD and QSAR datasets, it outperformed the other two cases in all metrics except minor Precision. Finally, with Cardiotocography dataset, it outperformed in GM, Kappa and minor Recall.

The results for the SVM classifier with all datasets that resulted in an increase in its performance using any metric are listed in

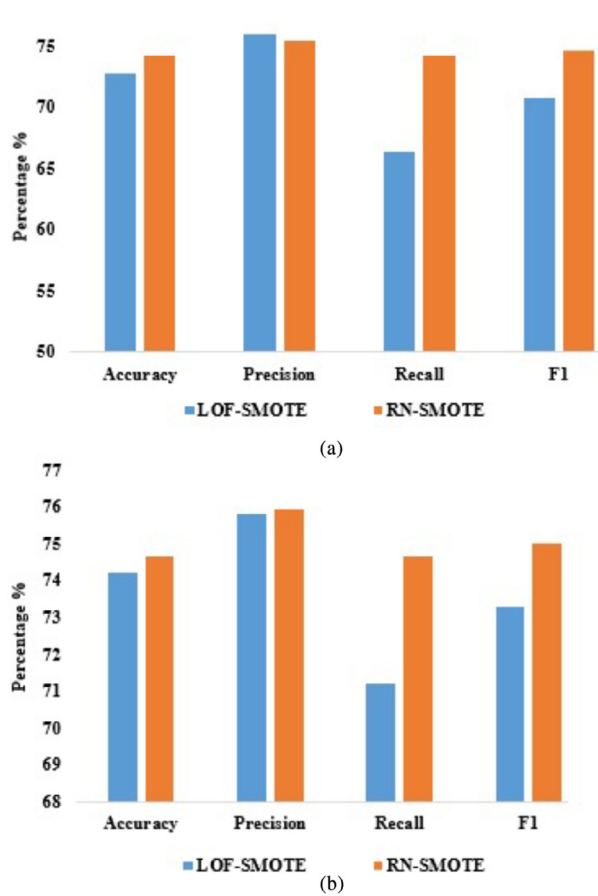


Fig. 9. RN-SMOTE vs LOF-SMOTE for PIMA dataset. (a) Naïve Bayes Classifier. (b) SVM Classifier.

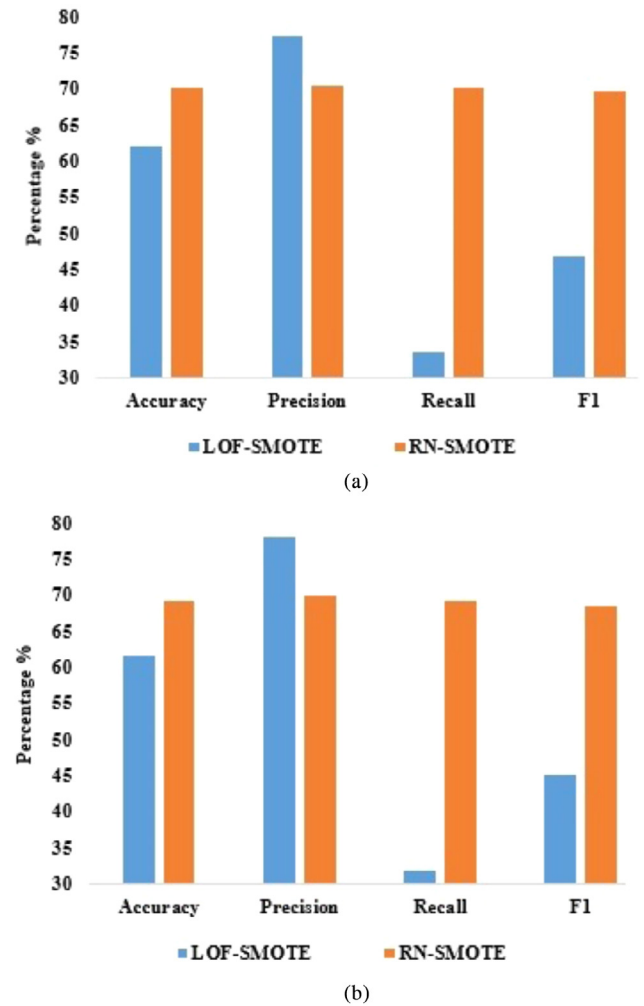


Fig. 10. RN-SMOTE vs LOF-SMOTE for Haberman dataset. (a) Naïve Bayes Classifier. (b) SVM Classifier.

Table 10. When trained and evaluated using InternetADS, WBC and Lymphography datasets the results show that, the performance of the SVM classifier combined with our **RN-SMOTE** outperformed the performance with each of the original datasets and the SMOTE. With all other listed datasets, the SVM had a hybrid performance. With Cardiocography and QSAR datasets it outperformed in all metrics except minor Precision for Cardiocography and except minor Recall for QSAR dataset. With Stamps dataset, it outperformed in GM, Kappa and minor Recall. Finally with Climate dataset, it outperformed in all metrics except GM and minor Recall.

The results for the Ada-Boost classifier with all datasets that resulted in an increase in its performance using all metrics are listed in **Table 11**. With all listed datasets, the Ada-Boost classifier combined with our **RN-SMOTE** had a hybrid performance. With InternetADS, Cardiocography, Lymphography, Climate and ILPD datasets it outperformed the other two cases in all metrics except minor Precision. With WBC dataset, it outperformed in all metrics except Kappa and minor Precision. While with Stamps dataset, it outperformed in GM, Kappa and minor Precision. Finally, with QSAR dataset, it outperformed in GM and minor F1 and Recall.

This section is the answer for RQ3 and RQ4. However, **Table 12** indicates more clearly which classifiers can perform well on real datasets with different dimensionalities and which of them can deal with datasets with critical imbalance ratios. RN-SMOTE improves the performance of RF, XGBOOST, SGD-LR, SVM and Ada-Boost on different dimensionalities datasets including Inter-

netADS which has 1557 features. Other classifiers cannot master this dimensionality variation. Also, RN-SMOTE enhances the performance of all classifiers on most of the real datasets with regular imbalanced ratios (31.22%– 50.92%) like Blood, ILPD and QSAR. In addition, RN-SMOTE improves the performance of XGBOOST, SGD-LR, SVM and Ada-Boost on most of the real datasets with critical imbalance ratios (1.94%– 2.25%) like Stamps, InternetADS, Cardiocography, WBC. Other classifiers can deal with only one or two of these challenging datasets.

4.6. Comparison with the Current State of Art

In this section, we compare the performance of RN-SMOTE with the performance of a current state of art namely SMOTE-LOF using the Glass, Haberman and PIMA datasets that are mainly used to evaluate SMOTE-LOF (Asniar et al., 2021). NB and SVM classifiers have been used for this comparison. The performance of the two classifiers is evaluated in terms of Accuracy, Precision, Recall and F1 scores where the classifiers are trained and evaluated using 5-fold cross validation. We included only the best among all results for the SMOTE-LOF model.

For PIMA dataset, the results proved that SVM classifier performance using Accuracy, Precision, Recall and F1 metrics with RN-SMOTE outperformed SMOTE-LOF by 0.46%, 0.15%, 3.48% and 1.7% respectively. Also, NB performance in terms of Accuracy,

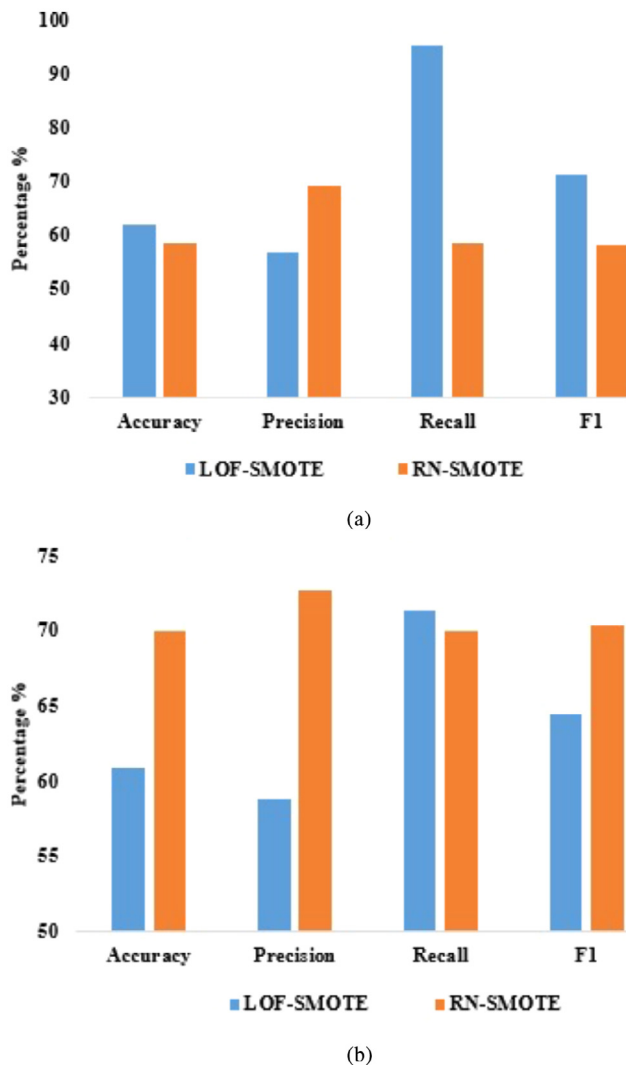


Fig. 11. RN-SMOTE vs LOF-SMOTE for Glass dataset. (a) Naïve Bayes Classifier. (b) SVM Classifier.

Recall and F1 with RN-SMOTE outperformed SMOTE-LOF by 1.56%, 7.85% and 3.85% while it has lagged by 0.53% in terms of Precision score. Fig. 9 draws a comparison between RN-SMOTE and SMOTE-LOF for PIMA dataset.

For Haberman dataset the results proved that, the performance of SVM classifier in terms of Accuracy, Recall and F1 with RN-SMOTE outperformed SMOTE-LOF by 7.6%, 37.41% and 23.28% respectively while it has lagged in Precision by 8.14%. Also, the performance of NB classifier in terms of Accuracy, Recall and F1 with RN-SMOTE outperformed SMOTE-LOF by 8.18%, 36.63% and 22.92% respectively while it has lagged in Precision by 6.82% (see Fig. 10.).

Lastly for the Glass dataset, the results proved that SVM classifier performance using Accuracy, Precision and F1 metrics with RN-SMOTE outperformed SMOTE-LOF by 9.07%, 13.95% and 5.83% respectively while using Recall SMOTE-LOF outperformed RN-SMOTE by 1.28%. Also, the results proved that NB classifier performance using Precision with RN-SMOTE outperformed SMOTE-LOF by 12.29% but using Accuracy, Recall and F1 SMOTE-LOF outperformed RN-SMOTE by 3.4%, 36.96% and 12.96% respectively. Fig. 11 draws a comparison between RN-SMOTE and SMOTE-LOF.

This section is the answer for RQ2. It is clear that RN-SMOTE can compete well with the recent method SMOTE-LOF. Keeping

in mind that LOF has estimated time complexity $O(n^2)$ while DBSCAN has estimated time complexity $O(n \log n)$ (Behera and Rani, 2016).

5. Conclusion and future work

This paper presented a model for imbalanced dataset classification. This model first oversamples the training data using SMOTE and then removes the noise from resulted oversampled data using DBSCAN which in turn is combined with the training data and introduced to SMOTE for balancing it. Based on the binary class datasets that have been used in the experiments, the results of evaluation experiments using six different metrics for imbalanced classification of 9 different classifiers using all datasets have proved that all classifiers have gained a notable improvement when combined with the proposed **RN-SMOTE** model compared to both SMOTE and original dataset. The performance improvement percentage is different from one dataset to another and from one classifier to another and from one metric to another. In addition, the results showed that KNN, NB, GB, and Ada-Boost classifiers resulted in a hybrid performance in terms of the used metrics. Also, other classifiers achieved high performance when combined with the proposed **RN-SMOTE** model in terms of all metrics for one or more datasets. Also, when compared to SMOTE-LOF, RN-SMOTE had an increase in the performance up to 9.07%, 13.95%, 37.41% and 23.28% in terms of Accuracy, Precision, Recall and F1 metrics.

As a future work, we will extend this work to include multiclass imbalanced datasets. Also, we plan to utilize deep learning model namely Generative Adversarial Networks (GAN) and other deep learning models for noise detection and removal and combine it with the different resampling techniques. We will also, utilize deep learning models for data resampling to overcome the limitations of the current techniques.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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